

The Collected Mathematical Papers of Arthur Cayley

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Papers [86] (cont.), [87], [88], [89], [90], [91]

[86] On the Developable derived from an Equation of the Fifth Order (*continued*)

[Book p. 504]

or the equations of the edge of regression are given by the system (equivalent of course to two equations),

$$\left\| \begin{array}{ccccc} \alpha, & \beta, & \gamma, & A, & B \\ \beta, & \gamma, & \delta, & B, & C \end{array} \right\| = 0.$$

The simplest mode of verifying *à posteriori* that the edge of regression is only of the ninth order, appears to be to consider this curve as the common intersection of the three surfaces of the seventh order:

$$A^3a - 3A^2Bb + 3AB^2c - B^3d = 0,$$

$$A^3b - 3A^2Bc + 3AB^2d - B^3e = 0,$$

$$A^3c - 3A^2Bd + 3AB^2e - B^3f = 0,$$

(which are at once obtained by combining the equation $Bt + A = 0$ with the cubic equations in t). It is obvious from a preceding equation that if the equations first given are multiplied by fA , $-2eA + fB$, $2dA - eB$, and added, an identical result is obtained. This shows that the curve of the forty-ninth order, the intersection of the first two surfaces, is made up of the curve in question, the curve of the fourth order $A = 0$, $B = 0$ (which reckons for thirty-six, as being a triple line on each surface), and the curve which is common to the two surfaces of the seventh order and the surface $2dA - eB = 0$. The equations of this last curve may be written,

$$e(af - 4be + 3cd) - 4d(bf - 4ce + 3d^2) = 0,$$

$$e^2a - 6e^2db + 12ed^2c - 8d^4 = 0,$$

$$e^2b - 6ed^2c + 4ed^3 = 0;$$

or, observing that these equations are

$$f(ae - 4bd) - 3c(be - 6ced + 4d^3) = 0,$$

$$e^2(ae - 4bd) - 2d(be^2 - 6ced + 4d^3) = 0,$$

$$e(be^2 - 6ced + 4d^3) = 0;$$

the last-mentioned curve is the intersection of

$$ae - 4bd = 0,$$

$$be^2 - 6ced + 4d^3 = 0,$$

where the second surface contains the double line $e = 0$, $d = 0$, which is also a single line upon the first surface. Omitting this extraneous line, the intersection is of the fourth order; and we may remark that, in passing, it is determined (exclusively of the double line) as the intersection of the three surfaces

$$ae - 4bd = 0,$$

$$be^2 - 6ced + 4d^3 = 0,$$

$$a^2d - 6abc + 4b^3 = 0,$$

[Book p. 505]

being in fact of the species iv. 4, of Mr Salmon's paper, "On the Classification of Curves of Double Curvature" [*Journal*, vol. v. (1850), pp. 23–46]. But returning to the question in hand; since $49 = 9 + 4 + 36$, we see

that the curve common to the three surfaces of the seventh order, or the edge of regression, is, as it ought to be, of the ninth order. It only remains to express the equation of the developable surface in terms of the functions $A, B, C, \alpha, \beta, \gamma, \delta$, which determine the stationary points and edge of regression; I have satisfied myself that the required formula is

$$\square = (AC - B^2)^2 - 1152 \{A(\beta\delta - \gamma^2) - B(\alpha\delta - \beta\gamma) + C(\alpha\gamma - \beta^2)\} = 0,$$

where the quantities $\alpha, \beta, \gamma, \delta$ may be replaced by their values in A, B, C ; and it will be noticed that when this is done, the terms of $\square$ are each of them at least of the third order in the last-mentioned functions.

I propose to term the family of developables treated of in this paper, ‘planar developables.’ In general, the coefficients of the generating plane of a developable being algebraical functions of a variable parameter t , the equation rationalized with respect to the parameter belongs to a system of n different planes; the developable which is the envelope of such a system may be termed a ‘multiplanar developable,’ and in the particular case of n being equal to unity, we have a planar developable. It would be very desirable to have some means of ascertaining from the equation of a developable what the degree of its ‘planarity’ is.

P.S.—At the time of writing the preceding paper I was under the impression that the only surface of the fourth order through the edge of regression was that given by the equation $AC - B^2 = 0$; but Mr Salmon has since made known to me an entirely new form of the function $\square$, the component functions of which, equated to zero, give six different surfaces of the fourth order, each of them passing through the edge of regression. The form in question is

$$3\square = LL' + 6MM' - 6NN',$$

where

$$\begin{aligned} L &= a^2f^2 + 225b^2e^2 - 32ace^2 - 32b^2df + 480bcde + 480c^3e - 84abef + 16acdf - 12bd^2f - 12ad^2e - 320bcde - 320c^2d^2, \\ L' &= 3a^2f^2 - 45b^2e^2 + 64ace^2 + 64b^2df - 12bcde - 80c^3e - 12abef - 12acdf - 84bc^2f - 8ad^2e + 20bcde. \end{aligned}$$

$$\begin{aligned} M &= 10bcd^2 - 18ad^3 - 15bc^2e + 32acde + 6b^2de - 9ac^2f + 2abdf + a^2ef - 8abe^2, \\ M' &= 10c^2de - 18cf^2 - 15bd^2e + 22bcd^2f + 6ad^3 - 9b^2ef + 2acef + abf^2 - 8ad^2f, \\ N &= 10b^3d^2 - 15b^2ce - 12acd^2 + 18ac^2e + abde - 2a^2b^2 + 6b^2f - 8abcf + 3a^2df, \\ N' &= 10c^2e^2 - 15bd^3 - 12c^2df + 18bd^2f + bcef - 2b^2f^2 + 6ae^2 - 8ade^2 + 3acf^2, \end{aligned}$$

where it should be noticed that

$$L + 3L' = -10(AC - B^2).$$

[Book p. 506]

The expressions of L, L', M, M', N, N' as linear functions of A, B, C (also due to Mr Salmon) are

$$\begin{aligned} L &= (11ae + 28bd - 39c^2)A + (af - 75be + 74cd)B + (11bf + 28ce - 39d^2)C, \\ L' &= (-7ae + 4bd + 3c^2)A + (3af + 10be - 13cd)B + (-7bf + 4ce + 3d^2)C, \\ M &= 3(bc - ad)A + 3(ae - c^2)B + (cd - af)C, \\ M' &= (cd - af)A + 3(bf - d^2)B + 3(de - cf)C, \\ N &= 3(b^2 - ac)A + 3(ad - bc)B + (bd - ae)C, \\ N' &= (ce - bf)A + 3(cf - de)B + 3(e^2 - df)C. \end{aligned}$$

I propose resuming the subject of these forms, and the general theory, in a subsequent paper. [This was never written.]

[87] Notes on Elliptic Functions (from Jacobi)

[Book p. 507]

[From the *Cambridge and Dublin Mathematical Journal*, vol. v. (1850), pp. 201–204.]

These Notes are mere translations from Jacobi's "Note sur une nouvelle application de l'analyse des fonctions elliptiques à l'algèbre," [*Crelle*, t. vii. (1831) pp. 41–43], and from the addition to the notice by him of the third supplement to Legendre's "Théorie des fonctions elliptiques" [*Crelle*, t. viii. (1832) pp. 413–417].

[88] On the Transformation of an Elliptic Integral

[Book p. 508]

[From the *Cambridge and Dublin Mathematical Journal*, vol. v. (1850), pp. 204–206.]

The following is a demonstration of a formula proved incidentally by Mr Boole (*Journal*, vol. ii. [1847] p. 7), in a paper "On the Attraction of a Solid of Revolution on an External Point."

Let

$$U = \int_{-1}^{+1} \frac{dx}{\sqrt{(1-x^2)\{1-(mx+n)^2\}}};$$

then, assuming

$$x = \frac{\alpha + iy}{1 - i\alpha y},$$

(so that $x = \pm 1$ gives $y = \pm 1$), we obtain

$$1 - x^2 = \frac{(1 + \alpha^2)(1 - y^2)}{(1 - i\alpha y)^2},$$

$$mx + n = \frac{(n - im\alpha) + (m - in\alpha)y}{1 - i\alpha y}.$$

Assume therefore

$$i\alpha + (n - im\alpha)(m - in\alpha) = 0,$$

whence

$$-i\alpha = \frac{(1 - m^2 - n^2) + \Delta}{2mn} \quad (\Delta^2 = 1 + m^4 + n^4 - 2m^2 - 2n^2 - 2m^2n^2).$$

[Book p. 509]

we find

$$1 - (mx + n)^2 = \frac{1 - (n - im\alpha)^2}{(1 - i\alpha y)^2} \{1 - (m - in\alpha)^2 y^2\},$$

and also

$$dx = \frac{(1 + \alpha^2) dy}{(1 - i\alpha y)^2},$$

whence

$$U = \sqrt{\left[\frac{1 + \alpha^2}{1 - (n - im\alpha)^2} \right]} \int_{-1}^{+1} \frac{dy}{\sqrt{(1 - y^2)\{1 - (m - in\alpha)^2 y^2\}}},$$

that is

$$U = 2\sqrt{\left[\frac{1 + \alpha^2}{1 - (n - im\alpha)^2} \right]} \int_0^1 \frac{dy}{\sqrt{(1 - y^2)\{1 - (m - in\alpha)^2 y^2\}}}.$$

But since

$$n - im\alpha = \frac{1 - m^2 + n^2 + \Delta}{2n},$$

$$m - in\alpha = \frac{1 + m^2 - n^2 + \Delta}{2m},$$

we have

$$1 - (n - im\alpha)^2 = -\frac{\Delta}{2n^2} (\Delta + 1 - m^2 + n^2),$$

$$1 + \alpha^2 = -\frac{\Delta}{2m^2n^2} (\Delta + 1 - m^2 - n^2);$$

and therefore

$$\frac{1 + \alpha^2}{1 - (n - im\alpha)^2} = \frac{1}{m^2} \cdot \frac{\Delta + 1 - m^2 - n^2}{\Delta + 1 - m^2 + n^2} = \frac{1}{m^2} \cdot \frac{(1 - m^2 - n^2 + \Delta)(1 - m^2 + n^2 - \Delta)}{(1 - m^2 + n^2 + \Delta)(1 - m^2 + n^2 - \Delta)} = \frac{2(1 + m^2 - n^2 + \Delta)}{4m^2};$$

consequently

$$U = \frac{1}{m} \sqrt{\{2(1 + m^2 - n^2 + \Delta)\}} \int_0^1 \frac{dy}{\sqrt{\left[(1 - y^2) \left\{1 - \frac{1 + m^2 - n^2 + \Delta}{2m} y^2\right\}\right]}}.$$

Write

$$\frac{1}{\lambda} = \frac{1 + m^2 - n^2 + \Delta}{2m}, \quad \lambda^2 k^2 = \frac{4mn}{(1 + m)^2 - n^2};$$

then

$$U = \frac{\sqrt{4\lambda k}}{\lambda} \cdot \frac{1}{\sqrt{\{(1 + m)^2 - n^2\}}} \int_0^1 \frac{dy}{\sqrt{(1 - y^2)(1 - \lambda^2 y^2)}},$$

where λ and k are connected by the relation that exists for the transformation of the second order, viz.

$$\lambda = \frac{2\sqrt{k}}{1 + k}.$$

[Book p. 510]

as may be immediately verified: hence, assuming

$$y = \frac{\sqrt{\lambda} z}{\sqrt{k} \sqrt{(1 - \lambda^2 z^2)}},$$

which gives

$$1 - y^2 = \frac{1 - z^2}{\sqrt{(1 - \lambda^2 z^2)}},$$

we find

$$U = \frac{4}{\sqrt{\{(1 + m)^2 - n^2\}}} \int_0^1 \frac{dz}{\sqrt{\left[(1 - z^2) \left\{1 - \frac{4mn}{(1 + m)^2 - n^2} z^2\right\}\right]}},$$

that is

$$\int_{-1}^{+1} \frac{dx}{\sqrt{(1 - x^2)\{1 - (mx + n)^2\}}} = 4 \int_0^1 \frac{dz}{\sqrt{(1 - z^2)\{(1 + m)^2 - n^2 - 4mn z^2\}}}.$$

Writing here

$$x = \cos \theta, \quad z = \cos \frac{1}{2} \phi,$$

then

$$\int_0^\pi \frac{d\theta}{\sqrt{1 - (m \cos \theta + n)^2}} = \int_0^\pi \frac{d\phi}{\sqrt{1 + m^2 - n^2 - 2mn \cos \phi}},$$

or if

$$m = \frac{r}{a}, \quad n = \frac{z}{a},$$

then finally

$$\int_0^\pi \frac{d\theta}{\sqrt{\{a^2 + (z + r \cos \theta)^2\}}} = \int_0^\pi \frac{d\phi}{\sqrt{(a^2 + r^2 + z^2 - 2ar \cos \phi)}},$$

the formula in question.

[89] On the Attraction of Ellipsoids (Jacobi's Method)

[Book p. 511]

[From the *Cambridge and Dublin Mathematical Journal*, vol. v. (1850), pp. 217–226.]

In a letter published in 1846 in Liouville's *Journal* (t. xi. p. 341) Jacobi says, "Il y a quatorze ans, je me suis posé le problème de chercher l'attraction d'un ellipsoïde homogène exercée sur un point extérieur quelconque par une méthode analogue à celle employée par Maclaurin par rapport aux points situés dans les axes principaux. J'y suis parvenu par trois substitutions consécutives. La première est une transformation de coordonnées; par la seconde le radical $\sqrt{(1 - m^2 \sin^2 \beta \cos^2 \psi - n^2 \sin^2 \beta \sin^2 \psi)}$ qui entre dans la double intégrale transformée est rendu rationnel au moyen de la double substitution

$$m \sin \beta \cos \psi = \sin v \cos \theta, \quad m \sin \beta \sin \psi = \sin v \sin \theta;$$

la troisième est encore une transformation de coordonnées. La recherche du sens géométrique de ces trois substitutions m'a conduit à approfondir la théorie des surfaces confocales par rapport auxquelles je découvris quantité de beaux théorèmes dont je communiquai quelques-uns des principaux à M. Steiner. Considérons l'ellipsoïde confocal mené par le point attiré P et le point p de l'ellipsoïde proposé, conjugué à P . Soient Q et q deux autres points conjugués quelconques situés respectivement sur l'ellipsoïde extérieur et intérieur. Menons de P un premier cône tangent à l'ellipsoïde intérieur, de p un second cône tangent à l'ellipsoïde extérieur. Ce dernier, tout imaginaire qu'il est, a ses trois axes réels (ainsi que ses deux droites focales). La première substitution ramène les axes de l'ellipsoïde à ceux du premier cône (c'est la substitution employée par Poisson, mais que j'avais antérieurement traitée et même étendue à un nombre quelconque de variables dans le mémoire *De binis Functionibus homogeneis &c.* [*Crelle*, t. xii. (1834) pp. 1–69]). Par la seconde substitution les angles que la droite Pq forme avec les axes du premier cône sont ramenés aux angles que la droite pQ forme avec les axes du second. Par la dernière substitution, on retourne de ces axes aux axes de l'ellipsoïde. La seconde substitution répond à un théorème de géométrie remarquable, savoir que 'Les cosinus des angles que la droite Pq forme avec deux des axes du premier cône sont en raison constante avec les cosinus des angles que la droite

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pQ forme avec deux des axes du second cône; ces deux axes sont les tangentes situés respectivement dans les sections de plus grande et de moindre courbure de chaque ellipsoïde, le troisième axe étant la normale à l'ellipsoïde.' Tout cela semble difficile à établir par la synthèse."

The object of this paper is to develop the above method of finding the attraction of an ellipsoid.

Consider an exterior ellipsoid, the squared semiaxes of which are $f + u$, $g + u$, $h + u$; and an interior ellipsoid, the squared semiaxes of which are $f + \bar{u}$, $g + \bar{u}$, $h + \bar{u}$. Let u , p , q be the elliptic coordinates of a point P on the exterior ellipsoid, the elliptic coordinates of the corresponding point $\bar{P}$ on the interior ellipsoid will be $\bar{u}$, p , q , and if a , b , c and $\bar{a}$, $\bar{b}$, $\bar{c}$ represent the ordinary coordinates of these points (the principal axes being the axes of coordinates), we have

$$\begin{aligned} a^2 &= \frac{(f+u)(f+q)(f+r)}{(f-g)(f-h)}, & \bar{a}^2 &= \frac{(f+\bar{u})(f+q)(f+r)}{(f-g)(f-h)}, \\ b^2 &= \frac{(g+u)(g+q)(g+r)}{(g-h)(g-f)}, & \bar{b}^2 &= \frac{(g+\bar{u})(g+q)(g+r)}{(g-h)(g-f)}, \\ c^2 &= \frac{(h+u)(h+q)(h+r)}{(h-f)(h-g)}, & \bar{c}^2 &= \frac{(h+\bar{u})(h+q)(h+r)}{(h-f)(h-g)}. \end{aligned}$$

I form the systems of equations

$$\begin{aligned} a_1^2 &= \frac{(u+f)(u+g)(u+h)}{(u-q)(u-r)}, & \bar{a}_1^2 &= \frac{(u+f)(u+g)(u+h)}{(u-q)(u-r)}, \\ b_1^2 &= \frac{(q+f)(q+g)(q+h)}{(q-r)(q-u)}, & \bar{b}_1^2 &= \frac{(q+f)(q+g)(q+h)}{(q-r)(q-\bar{u})}, \\ c_1^2 &= \frac{(r+f)(r+g)(r+h)}{(r-u)(r-q)}, & \bar{c}_1^2 &= \frac{(r+f)(r+g)(r+h)}{(r-\bar{u})(r-q)}. \end{aligned}$$

$$\begin{aligned}
 \alpha &= \frac{a_1 a}{f + u}, & \beta &= \frac{a_1 b}{g + u}, & \gamma &= \frac{a_1 c}{h + u}, \\
 \alpha' &= \frac{b_1 a}{f + q}, & \beta' &= \frac{b_1 b}{g + q}, & \gamma' &= \frac{b_1 c}{h + q}, \\
 \alpha'' &= \frac{c_1 a}{f + r}, & \beta'' &= \frac{c_1 b}{g + r}, & \gamma'' &= \frac{c_1 c}{h + r}, \\
 \bar{\alpha} &= \frac{\bar{a}_1 \bar{a}}{f + \bar{u}}, & \bar{\beta} &= \frac{\bar{a}_1 \bar{b}}{g + \bar{u}}, & \bar{\gamma} &= \frac{\bar{a}_1 \bar{c}}{h + \bar{u}}, \\
 \bar{\alpha}' &= \frac{\bar{b}_1 \bar{a}}{f + q}, & \bar{\beta}' &= \frac{\bar{b}_1 \bar{b}}{g + q}, & \bar{\gamma}' &= \frac{\bar{b}_1 \bar{c}}{h + q}, \\
 \bar{\alpha}'' &= \frac{\bar{c}_1 \bar{a}}{f + r}, & \bar{\beta}'' &= \frac{\bar{c}_1 \bar{b}}{g + r}, & \bar{\gamma}'' &= \frac{\bar{c}_1 \bar{c}}{h + r}.
 \end{aligned}$$

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And then writing

$$\begin{aligned}
 X &= \alpha X_1 + \alpha' Y_1 + \alpha'' Z_1, & \bar{X} &= \bar{\alpha} \bar{X}_1 + \bar{\alpha}' \bar{Y}_1 + \bar{\alpha}'' \bar{Z}_1, \\
 Y &= \beta X_1 + \beta' Y_1 + \beta'' Z_1, & \bar{Y} &= \bar{\beta} \bar{X}_1 + \bar{\beta}' \bar{Y}_1 + \bar{\beta}'' \bar{Z}_1, \\
 Z &= \gamma X_1 + \gamma' Y_1 + \gamma'' Z_1, & \bar{Z} &= \bar{\gamma} \bar{X}_1 + \bar{\gamma}' \bar{Y}_1 + \bar{\gamma}'' \bar{Z}_1,
 \end{aligned}$$

if X, Y, Z are the cosines of the inclinations of a line PQ to the principal axes of the ellipsoids, X_1, Y_1, Z_1 will be the cosines of the inclinations of this line to the principal axes of the cone having P for its vertex, and circumscribed about the interior ellipsoid. In like manner, $\bar{X}, \bar{Y}, \bar{Z}$ being the cosines of the inclinations of a line $\bar{P}\bar{Q}$ to the principal axes of the ellipsoids, $\bar{X}_1, \bar{Y}_1, \bar{Z}_1$ will be the cosines of the inclinations of this line to the principal axes of the cone having $\bar{P}$ for its vertex and circumscribed about the exterior ellipsoid. Assuming that the points $Q, \bar{Q}$ are situated upon the exterior and interior ellipsoids respectively, suppose that X_1, Y_1, Z_1 and $\bar{X}_1, \bar{Y}_1, \bar{Z}_1$ are connected by the equivalent systems of equations,

$$\begin{aligned}
 X_1 &= \sqrt{(u - \bar{u})} \sqrt{\left(\frac{\bar{X}_1^2}{u - \bar{u}} + \frac{\bar{Y}_1^2}{\bar{u} - q} + \frac{\bar{Z}_1^2}{\bar{u} - r} \right)}, \\
 Y_1 &= \sqrt{\left(\frac{\bar{u} - q}{u - q} \right)} \bar{Y}_1, & Z_1 &= \sqrt{\left(\frac{\bar{u} - r}{u - r} \right)} \bar{Z}_1, \\
 \bar{X}_1 &= \sqrt{(u - \bar{u})} \sqrt{\left(\frac{X_1^2}{u - \bar{u}} + \frac{Y_1^2}{u - q} + \frac{Z_1^2}{u - r} \right)}, \\
 \bar{Y}_1 &= \sqrt{\left(\frac{u - q}{\bar{u} - q} \right)} Y_1, & \bar{Z}_1 &= \sqrt{\left(\frac{u - r}{\bar{u} - r} \right)} Z_1,
 \end{aligned}$$

then it will presently be shown that the points $Q, \bar{Q}$ are corresponding points, which will prove the geometrical theorem of Jacobi. Before proceeding further it will be convenient to notice the formulae

$$1 - \frac{a^2}{f + u} - \frac{b^2}{g + u} - \frac{c^2}{h + u} = \frac{\bar{u} - u}{a_1^2},$$

$$\frac{Xa}{f + u} + \frac{Yb}{g + u} + \frac{Zc}{h + u} = \frac{\bar{u} - u}{a_1^2} \left(\frac{X_1 a_1}{u - \bar{u}} + \frac{Y_1 b_1}{u - q} + \frac{Z_1 c_1}{u - r} \right),$$

$$\left(\frac{Xa}{f + u} + \frac{Yb}{g + u} + \frac{Zc}{h + u} \right)^2 + \frac{\bar{u} - u}{a_1^2} \left(\frac{X^2}{f + u} + \frac{Y^2}{g + u} + \frac{Z^2}{h + u} \right) = \frac{\bar{u} - u}{a_1^2} \left(\frac{X_1^2}{u - \bar{u}} + \frac{Y_1^2}{u - q} + \frac{Z_1^2}{u - r} \right).$$

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and the corresponding ones

$$1 - \frac{\bar{a}^2}{f + \bar{u}} - \frac{\bar{b}^2}{g + \bar{u}} - \frac{\bar{c}^2}{h + \bar{u}} = \frac{u - \bar{u}}{\bar{a}_1^2},$$

$$\frac{\bar{X}\bar{a}}{f + \bar{u}} + \frac{\bar{Y}\bar{b}}{g + \bar{u}} + \frac{\bar{Z}\bar{c}}{h + \bar{u}} = \frac{u - \bar{u}}{\bar{a}_1^2} \left(\frac{\bar{X}_1\bar{a}_1}{\bar{u} - u} + \frac{\bar{Y}_1\bar{b}_1}{\bar{u} - q} + \frac{\bar{Z}_1\bar{c}_1}{\bar{u} - r} \right),$$

$$\left(\frac{\bar{X}\bar{a}}{f + \bar{u}} + \frac{\bar{Y}\bar{b}}{g + \bar{u}} + \frac{\bar{Z}\bar{c}}{h + \bar{u}} \right)^2 + \frac{u - \bar{u}}{\bar{a}_1^2} \left(\frac{\bar{X}^2}{f + \bar{u}} + \frac{\bar{Y}^2}{g + \bar{u}} + \frac{\bar{Z}^2}{h + \bar{u}} \right) = \frac{u - \bar{u}}{\bar{a}_1^2} \left(\frac{\bar{X}_1^2}{\bar{u} - u} + \frac{\bar{Y}_1^2}{\bar{u} - q} + \frac{\bar{Z}_1^2}{\bar{u} - r} \right).$$

The coordinates of the point Q are obviously $a + \rho X$, $b + \rho Y$, $c + \rho Z$ (where $\rho = PQ$); substituting these values in the equation of the interior ellipsoid, we obtain

$$\rho \left\{ \frac{(u - \bar{u})}{\bar{a}_1^2} + 2\rho \left(\frac{Xa}{f + \bar{u}} + \frac{Yb}{g + \bar{u}} + \frac{Zc}{h + \bar{u}} \right) + \rho^2 \left(\frac{X^2}{f + \bar{u}} + \frac{Y^2}{g + \bar{u}} + \frac{Z^2}{h + \bar{u}} - 1 \right) \right\} = 0;$$

reducing the coefficients of this equation by the formulae first given, and omitting a factor $\frac{\bar{u} - u}{\bar{a}_1^2}$, we obtain

$$\frac{\bar{a}_1^2 \bar{X}_1^2}{(\bar{u} - u)^2} - \left(\frac{X_1 a_1}{\bar{u} - u} - \frac{Y_1 b_1}{\bar{u} - q} + \frac{Z_1 c_1}{\bar{u} - r} \right)^2 \rho^2 + 2\rho \left(\frac{X_1 a_1}{\bar{u} - u} + \frac{Y_1 b_1}{\bar{u} - q} + \frac{Z_1 c_1}{\bar{u} - r} \right) - 1 = 0,$$

that is,

$$\frac{\bar{a}_1^2 \bar{X}_1^2}{(\bar{u} - u)^2} \rho^2 = \left\{ \rho \left(\frac{X_1 a_1}{\bar{u} - u} - \frac{Y_1 b_1}{\bar{u} - q} + \frac{Z_1 c_1}{\bar{u} - r} \right) - 1 \right\}^2,$$

or

$$\rho = \frac{1}{\frac{X_1 a_1 - X_1 \bar{a}_1}{\bar{u} - u} + \frac{Y_1 b_1}{\bar{u} - q} + \frac{Z_1 c_1}{\bar{u} - r}},$$

which is easily transformed into

$$\rho = \frac{\bar{u} - u}{a_1 X_1 - \bar{a}_1 \bar{X}_1 + \frac{(f + u)b_1 Y_1 - (f + \bar{u})\bar{b}_1 \bar{Y}_1}{f + q} + \frac{(f + u)c_1 Z_1 - (f + \bar{u})\bar{c}_1 \bar{Z}_1}{f + r}};$$

and this form remaining unaltered when u and $\bar{u}$ are interchanged, it follows that if $PQ = \rho$, then $\rho = \bar{\rho}$, which is a known theorem. The value of ρ or $\bar{\rho}$ may however be expressed in a yet simpler form; for, considering the expression

$$\begin{aligned} \frac{X}{\sqrt{(f + u)}} - \frac{\bar{X}}{\sqrt{(f + \bar{u})}} &= \frac{a}{\sqrt{(f + u)}} \left\{ \frac{a_1 X_1}{f + u} + \frac{b_1 Y_1}{f + q} + \frac{c_1 Z_1}{f + r} \right\} - \frac{\bar{a}}{\sqrt{(f + \bar{u})}} \left\{ \frac{\bar{a}_1 \bar{X}_1}{f + \bar{u}} + \frac{\bar{b}_1 \bar{Y}_1}{f + q} + \frac{\bar{c}_1 \bar{Z}_1}{f + r} \right\} \\ &= \frac{-1}{\bar{u} - u} \left\{ \frac{a}{\sqrt{(f + u)}} - \frac{\bar{a}}{\sqrt{(f + \bar{u})}} \right\} \\ &\times \left\{ a_1 X_1 - \bar{a}_1 \bar{X}_1 + \frac{(f + u)b_1 Y_1 - (f + \bar{u})\bar{b}_1 \bar{Y}_1}{f + q} + \frac{(f + u)c_1 Z_1 - (f + \bar{u})\bar{c}_1 \bar{Z}_1}{f + r} \right\}; \end{aligned}$$

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we see that

$$\frac{X}{\sqrt{(f + u)}} - \frac{\bar{X}}{\sqrt{(f + \bar{u})}} = -\frac{1}{\rho} \left\{ \frac{a}{\sqrt{(f + u)}} - \frac{\bar{a}}{\sqrt{(f + \bar{u})}} \right\},$$

and similarly

$$\begin{aligned} \frac{Y}{\sqrt{(g + u)}} - \frac{\bar{Y}}{\sqrt{(g + \bar{u})}} &= -\frac{1}{\rho} \left\{ \frac{b}{\sqrt{(g + u)}} - \frac{\bar{b}}{\sqrt{(g + \bar{u})}} \right\}, \\ \frac{Z}{\sqrt{(h + u)}} - \frac{\bar{Z}}{\sqrt{(h + \bar{u})}} &= -\frac{1}{\rho} \left\{ \frac{c}{\sqrt{(h + u)}} - \frac{\bar{c}}{\sqrt{(h + \bar{u})}} \right\}; \end{aligned}$$

which are in fact the equations which express that Q and $\bar{Q}$ are corresponding points.

It is proper to remark that supposing, as we are at liberty to do, that $P, \bar{P}$ are situate in corresponding octants of the two ellipsoids, then if the curve of contact of the circumscribed cone having P for its vertex divide the surface of the interior ellipsoid into two parts $\bar{M}, \bar{N}$, of which the former lies contiguous to $\bar{P}$: also if the curve of intersection of the tangent plane at $\bar{P}$ divide the surface of the exterior ellipsoid into two parts M, N , of which M lies contiguous to the point P ; then the different points of $M, \bar{M}$ correspond to each other, as do also the different points of $N, \bar{N}$.

We may now pass to the integral calculus problem. The Attraction parallel to the axis of x is

$$A = \int \frac{x \, dx \, dy \, dz}{(x^2 + y^2 + z^2)^{\frac{3}{2}}},$$

the limits of the integration being given ($\bar{a}^2$)

$$\frac{x^2}{f + \bar{u}} + \frac{y^2}{g + \bar{u}} + \frac{z^2}{h + \bar{u}} = 1;$$

or putting

$$x = rX, \quad y = rY, \quad z = rZ,$$

where X, Y, Z have the same signification as before, we have

$$dx \, dy \, dz = r^2 \, dr \, dS,$$

and then

$$A = \int X \, dr \, dS = \int \rho X \, dS,$$

where ρ has the same signification as before: it will be convenient to leave the formula in this form, rather than to take at once the difference of the two values of ρ , but of course the integration is as in the ordinary methods to be performed so as to extend to the whole volume of the ellipsoid. The expression dS denotes the differen-

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tial of a spherical surface radius unity, and if θ, ϕ are the parameters by which the position of ρ is determined, we have

$$dS = \begin{vmatrix} X, & Y, & Z \\ \frac{dX}{d\theta}, & \frac{dY}{d\theta}, & \frac{dZ}{d\theta} \\ \frac{dX}{d\phi}, & \frac{dY}{d\phi}, & \frac{dZ}{d\phi} \end{vmatrix} d\theta \, d\phi.$$

In the present case

$$dS = dS_1 = \begin{vmatrix} X_1, & Y_1, & Z_1 \\ \frac{dX_1}{dY_1}, & \frac{dY_1}{dY_1}, & \frac{dZ_1}{dY_1} \\ \frac{dX_1}{dZ_1}, & \frac{dY_1}{dZ_1}, & \frac{dZ_1}{dZ_1} \end{vmatrix} dY_1 \, dZ_1.$$

or from the values of X_1, Y_1, Z_1 in terms of $\bar{X}_1, \bar{Y}_1, \bar{Z}_1$ (observing that X_1 must be replaced by its value $\sqrt{(1 - Y_1^2 - Z_1^2)}$) we deduce

$$dS = \sqrt{\left[\frac{(\bar{u} - q)(\bar{u} - r)}{(u - q)(u - r)} \right]} \frac{1}{\bar{X}_1} d\bar{Y}_1 \, d\bar{Z}_1.$$

But $d\bar{S} = d\bar{S}_1 = \frac{1}{\bar{X}_1} d\bar{Y}_1 \, d\bar{Z}_1$, whence

$$dS = \sqrt{\left[\frac{(\bar{u} - q)(\bar{u} - r)}{(u - q)(u - r)} \right]} \frac{\bar{X}_1 \, d\bar{S}}{X_1},$$

which shows that the corresponding elements of the spheres whose centres are $P, \bar{P}$, projected upon the tangent planes at P and $\bar{P}$ respectively, are in a constant ratio. It may be noticed also that if $\mu, \bar{\mu}$ are the masses of the ellipsoids, the ratio in question

$$= \sqrt{\left[\frac{(\bar{u} - q)(\bar{u} - r)}{(u - q)(u - r)} \right]} = \frac{\bar{\mu} \bar{a}_1}{\mu \bar{a}_1}.$$

We have thus

$$A = \sqrt{\left[\frac{(\bar{u} - q)(\bar{u} - r)}{(u - q)(u - r)} \right]} \int \frac{\bar{\rho} \bar{X}_1 X d\bar{S}}{X_1},$$

that is

$$A = \sqrt{\left[\frac{(\bar{u} - q)(\bar{u} - r)}{(u - q)(u - r)} \right]} \int \frac{\bar{\rho} (\alpha X_1 + \alpha' Y_1 X_1 + \alpha'' Z_1) X_1 d\bar{S}}{X_1}.$$

The value which it will be convenient to use for ρ is that derived from the equation

$$\bar{\rho}^2 \left(\frac{X^2}{f + \bar{u}} + \frac{Y^2}{g + \bar{u}} + \frac{Z^2}{h + \bar{u}} \right) + 2\bar{\rho} \left(\frac{\bar{X}\bar{a}}{f + \bar{u}} + \frac{\bar{Y}\bar{b}}{g + \bar{u}} + \frac{\bar{Z}\bar{c}}{h + \bar{u}} \right) + \frac{\bar{u} - u}{\bar{a}_1^2} = 0.$$

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with only the transformation of expressing the radical in terms of X_1 , viz.

$$\bar{\rho} = \frac{-\frac{\bar{X}\bar{a}}{f + \bar{u}} - \frac{\bar{Y}\bar{b}}{g + \bar{u}} - \frac{\bar{Z}\bar{c}}{h + \bar{u}} + \frac{1}{\bar{a}_1} X_1}{\frac{X^2}{f + \bar{u}} + \frac{Y^2}{g + \bar{u}} + \frac{Z^2}{h + \bar{u}}};$$

substituting these values and observing that Y_1 and Z_1 are rational functions of X, Y , and Z , but that X_1 is a radical, and that in order to extend the integration to the whole ellipsoid, the values corresponding to the opposite signs of X_1 will require to be added, the quantity to be integrated (omitting for the moment the exterior constant factor) is

$$\frac{-\left(\frac{\bar{X}\bar{a}}{f + \bar{u}} + \frac{\bar{Y}\bar{b}}{g + \bar{u}} + \frac{\bar{Z}\bar{c}}{h + \bar{u}} \right) \frac{1}{\bar{a}_1} \{ \alpha' Y_1 + \alpha'' Z_1 \} X_1 d\bar{S}}{\frac{X^2}{f + \bar{u}} + \frac{Y^2}{g + \bar{u}} + \frac{Z^2}{h + \bar{u}}},$$

the integration to be extended over the spherical area $\bar{S}$. Consider the quantity within $\{ \}$, this is

$$\alpha \left(\frac{\bar{X}\bar{a}}{f + \bar{u}} + \frac{\bar{Y}\bar{b}}{g + \bar{u}} + \frac{\bar{Z}\bar{c}}{h + \bar{u}} \right) + \frac{\alpha'}{\bar{a}_1} \sqrt{\left(\frac{\bar{u} - q}{u - q} \right)} (\bar{\alpha}' \bar{X} + \bar{\beta}' \bar{Y} + \bar{\gamma}' \bar{Z}) + \frac{\alpha''}{\bar{a}_1} \sqrt{\left(\frac{\bar{u} - r}{u - r} \right)} (\bar{\alpha}'' \bar{X} + \bar{\beta}'' \bar{Y} + \bar{\gamma}'' \bar{Z}).$$

The coefficients of $\bar{Y}$ and $\bar{Z}$ vanish, in fact that of $\bar{Y}$ is

$$\begin{aligned} & \frac{a_1 a}{f + u} \frac{\bar{b}}{g + \bar{u}} + \frac{b_1 a}{a_1 (f + q)} \sqrt{\left(\frac{\bar{u} - q}{u - q} \right)} \frac{\bar{b}_1 \bar{b}}{g + q} + \frac{c_1 a}{a_1 (f + r)} \sqrt{\left(\frac{\bar{u} - r}{u - r} \right)} \frac{\bar{c}_1 \bar{b}}{g + r} \\ &= \frac{a \bar{b}}{a_1} \left\{ \frac{a_1^2}{(f + u)(g + \bar{u})} + \frac{b_1 \bar{b}_1}{(f + q)(g + q)} \sqrt{\left(\frac{\bar{u} - q}{u - q} \right)} + \frac{c_1 \bar{c}_1}{(f + r)(g + r)} \sqrt{\left(\frac{\bar{u} - r}{u - r} \right)} \right\} \\ &= \frac{a \bar{b}}{a_1} \left\{ \frac{a_1^2}{(f + u)(g + u)} + \frac{b_1^2}{(f + q)(g + q)} + \frac{c_1^2}{(f + r)(g + r)} \right\} = 0, \end{aligned}$$

and similarly for the coefficient of $\bar{Z}$.

The coefficient of $\bar{X}$ is in like manner shown to be

$$\frac{a\bar{a}}{a_1} \left\{ \frac{a_1^2}{(f+u)^2} + \frac{b_1^2}{(f+q)^2} + \frac{c_1^2}{(f+r)^2} \right\} = \frac{a\bar{a}}{a_1} \cdot \frac{(f-g)(f-h)}{(f+u)(f+q)(f+r)} = \frac{\bar{a}}{a^2 a_1} = \frac{\bar{a}}{aa'_1},$$

or the quantity in question is simply

$$\frac{\bar{a}}{aa_1} \bar{X}.$$

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Multiplying this by $\bar{X}_1 = \bar{\alpha}\bar{X} + \bar{\alpha}'\bar{Y} + \bar{\alpha}''\bar{Z}$, the terms containing $\bar{X}\bar{Y}$, $\bar{X}\bar{Z}$ vanish after the integration, and we need only consider the term $\bar{\alpha}\bar{X}^2$, or what is the same $\frac{\bar{a}^2\bar{a}_1}{aa_1(f+\bar{u})} \bar{X}^2$. Hence

$$A = \sqrt{\left[\frac{(\bar{u}-q)(\bar{u}-r)}{(u-q)(u-r)} \right]} \frac{\bar{a}^2\bar{a}_1}{aa_1(f+\bar{u})} \int \frac{\bar{X}^2 d\bar{S}}{\frac{\bar{X}^2}{f+\bar{u}} + \frac{\bar{Y}^2}{g+\bar{u}} + \frac{\bar{Z}^2}{h+\bar{u}}}.$$

The value of the corresponding function $\bar{A}$ (that is, the attraction of the exterior ellipsoid upon $\bar{P}$) is

$$\bar{A} = \frac{\bar{a}}{f+\bar{u}} \int \frac{\bar{X}^2 d\bar{S}}{\frac{\bar{X}^2}{f+\bar{u}} + \frac{\bar{Y}^2}{g+\bar{u}} + \frac{\bar{Z}^2}{h+\bar{u}}},$$

the limits being the same, whence

$$\frac{A}{\bar{A}} = \sqrt{\left[\frac{(\bar{u}-q)(\bar{u}-r)}{(u-q)(u-r)} \right]} \frac{f+\bar{u}}{f+u} \cdot \frac{\bar{a}a_1}{a\bar{a}_1} = \frac{\sqrt{(u+g)}\sqrt{(u+h)}}{\sqrt{(\bar{u}+g)}\sqrt{(\bar{u}+h)}},$$

or we have

$$A = \frac{\sqrt{(u+g)}\sqrt{(u+h)}}{\sqrt{(\bar{u}+g)}\sqrt{(\bar{u}+h)}} \bar{A}, \quad B = \frac{\sqrt{(u+h)}\sqrt{(u+f)}}{\sqrt{(\bar{u}+h)}\sqrt{(\bar{u}+f)}} \bar{B}, \quad C = \frac{\sqrt{(u+f)}\sqrt{(u+g)}}{\sqrt{(\bar{u}+f)}\sqrt{(\bar{u}+g)}} \bar{C},$$

formulae which constitute in fact Ivory's theorem.

Let K , $\bar{K}$ denote the attractions in the directions of the normals at P , $\bar{P}$, we have

$$K = \int X d\bar{S}, \quad \bar{K} = \int \bar{X} d\bar{S},$$

or

$$K = \frac{\bar{\mu}a_1}{\mu\bar{a}_1} \bar{K};$$

and it is important to remark that this is true not only for the entire ellipsoids; but if $\mathfrak{M}$, $\mathfrak{N}$ denote the attractions of the cones standing on the portions $\bar{M}$, $\bar{N}$ of the surface of the interior ellipsoid, and $\bar{\mathfrak{M}}$, $\bar{\mathfrak{N}}$ the attractions of the portions of the exterior ellipsoid bounded by the tangent plane at $\bar{P}$, and the portions M , N of the surface of the exterior ellipsoid, then

$$\mathfrak{M} = \frac{\bar{\mu}a_1}{\mu\bar{a}_1} \bar{\mathfrak{M}}, \quad \mathfrak{N} = \frac{\bar{\mu}a_1}{\mu\bar{a}_1} \bar{\mathfrak{N}},$$

where obviously

$$K = \mathfrak{M} - \mathfrak{N}, \quad \bar{K} = \bar{\mathfrak{M}} + \bar{\mathfrak{N}};$$

this theorem is so far as I am aware new.

[90] Note sur quelques formules relatives aux coniques

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[From the *Journal für die reine und angewandte Mathematik (Crelle)*, tom. XXXIX. (1850), pp. 1–3.]

Soit, comme à l'ordinaire:

$$\begin{aligned}\mathfrak{A} &= BC - F^2, \\ \mathfrak{B} &= CA - G^2, \\ \mathfrak{C} &= AB - H^2, \\ \mathfrak{F} &= GH - AF, \\ \mathfrak{G} &= HF - BG, \\ \mathfrak{H} &= FG - CH, \\ K &= ABC - AF^2 - BG^2 - CH^2 + 2FGH,\end{aligned}\dots\dots(1)$$

et désignons, pour abréger, les fonctions

$$\begin{aligned}Ax^2 + By^2 + Cz^2 + 2Fyz + 2Gxz + 2Hxy, \\ A\alpha x + B\beta y + C\gamma z + F(\beta z + \gamma y) + G(\gamma x + \alpha z) + H(\alpha y + \beta x), \\ A(\beta z - \gamma y)^2 + B(\gamma x - \alpha z)^2 + C(\alpha y - \beta x)^2 + 2F(\gamma x - \alpha z)(\alpha y - \beta x) \\ + 2G(\alpha y - \beta x)(\beta z - \gamma y) + 2H(\beta z - \gamma y)(\gamma x - \alpha z); \\ \&c.\end{aligned}$$

par

$$Ax^2 + \dots; \quad A\alpha x + \dots; \quad A(\beta z - \gamma y)^2 + \dots; \quad \&c.$$

Cela posé, soient $\mathfrak{A} + \mathfrak{a}$, $\mathfrak{B} + \mathfrak{b}$, $\mathfrak{C} + \mathfrak{c}$, $\mathfrak{F} + \mathfrak{f}$, $\mathfrak{G} + \mathfrak{g}$, $\mathfrak{H} + \mathfrak{h}$, $K + k$ ce que deviennent $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{F}$, $\mathfrak{G}$, $\mathfrak{H}$, K , en écrivant $A + \alpha^2$, $B + \beta^2$, $C + \gamma^2$, $F + \beta\gamma$, $G + \gamma\alpha$, $H + \alpha\beta$ au lieu de A , B , C , F , G , H . Alors on aura d'abord

$$k = \mathfrak{A}\alpha^2 + \dots \dots\dots(2)$$

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et les quantités $\mathfrak{a}$, $\mathfrak{b}$, $\mathfrak{c}$, $\mathfrak{f}$, $\mathfrak{g}$, $\mathfrak{h}$ seront données par l'équation

$$\mathfrak{a}\alpha^2 + \dots = A(\beta z - \gamma y)^2 + \dots \dots\dots(3),$$

ou, si l'on veut, par celle-ci:

$$\mathfrak{a}\alpha x_1 + \dots = A(\beta z - \gamma y)(\beta z_1 - \gamma y_1) + \dots \dots\dots(4),$$

(savoir, en considérant ces équations comme identiques par rapport à x , y , z et x_1 , y_1 , z_1 , respectivement).

On obtient sans difficulté les équations identiques:

$$(A\alpha x_1 + \dots)(Axx_1 + \dots) - (A\alpha x + \dots)(A\alpha_1 x_1 + \dots) = \mathfrak{A}(\beta z_1 - \gamma y_1)(\beta z - \gamma y) + \dots \quad (5),$$

$$(\mathfrak{a}\alpha\alpha_1 + \dots)(\mathfrak{a}xx_1 + \dots) - (\mathfrak{a}\alpha x + \dots)(\mathfrak{a}\alpha_1 x_1 + \dots) = -K [A(\beta z_1 - \gamma y_1)(\beta z - \gamma y) + \dots] \quad (6).$$

Comme on sait, la condition sous laquelle la droite $lx + my + nz = 0$ touche la conique $U = Ax^2 + \dots = 0$ peut être présentée sous la forme $\mathfrak{U}^2 + \dots = 0$. Donc la condition pour que cette droite touche la conique $U + (\alpha x + \beta y + \gamma z)^2 = 0$, est

$$\mathfrak{U}^2 + \dots + A(\gamma m - \beta n)^2 + \dots = 0 \dots\dots\dots(7).$$

En réduisant au moyen de l'équation $(\mathfrak{A}\alpha^2 + \dots)(\mathfrak{U}^2 + \dots) = K [A(\gamma m - \beta n)^2 + \dots]$ (laquelle n'est qu'un cas particulier de l'équation (6)), cette condition devient:

$$(K + \mathfrak{A}\alpha^2 + \dots)(\mathfrak{U}^2 + \dots) - (\mathfrak{A}\alpha l + \dots)^2 = 0 \dots\dots\dots(8).$$

Pour trouver la condition sous laquelle les coniques

$$U + (\alpha x + \beta y + \gamma z)^2 = 0, \quad U + (\alpha_1 x + \beta_1 y + \gamma_1 z)^2 = 0$$

se touchent, on n'a qu'à remarquer que l'équation de la tangente commune est, ou

$$(\alpha - \alpha_1)x + (\beta - \beta_1)y + (\gamma - \gamma_1)z = 0, \quad \text{ou} \quad (\alpha + \alpha_1)x + (\beta + \beta_1)y + (\gamma + \gamma_1)z = 0.$$

En ne considérant que la première de ces droites, on a pour la condition cherchée:

$$\mathfrak{A}(\alpha - \alpha_1)^2 + \dots + A(\beta\gamma_1 - \beta_1\gamma)^2 + \dots = 0 \quad \dots\dots\dots (9);$$

ou, en réduisant au moyen du même cas particulier de l'équation (6), on obtient cette condition sous la forme

$$(K + \mathfrak{A}\alpha^2 + \dots)(K + \mathfrak{A}\alpha_1^2 + \dots) - (K + \mathfrak{A}\alpha\alpha_1 + \dots)^2 = 0 \quad \dots\dots\dots (10).$$

On sait que les coordonnées X, Y, Z du pôle de la droite $lx + my + nz = 0$, par rapport à la conique $U = 0$, peuvent être trouvées par l'équation

$$\lambda X + \mu Y + \nu Z = \mathfrak{A}l\lambda + \dots,$$

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(considérée comme identique par rapport à λ, μ, ν). De là les coordonnées X, Y, Z du pôle de cette même droite par rapport à la conique

$$U + (\alpha x + \beta y + \gamma z)^2 = 0$$

se trouveront par l'équation

$$\lambda X + \mu Y + \nu Z = \mathfrak{A}l\lambda + \dots + A(\gamma m - \beta n)(\gamma \mu - \beta \nu). \quad (11)$$

(considérée comme identique par rapport à λ, μ, ν). Cette équation peut aussi être présentée sous la forme

$$K(\lambda X + \mu Y + \nu Z) = (K + \mathfrak{A}\alpha^2 + \dots)(\mathfrak{A}l\lambda + \dots) - (\mathfrak{A}\alpha\lambda + \dots)(\mathfrak{A}\alpha l + \dots) \quad \dots\dots (12),$$

ce qui peut être démontré facilement au moyen d'un cas particulier de l'équation (6).

Si les deux droites $lx + my + nz = 0, l'x + m'y + n'z = 0$, touchent la conique $U = 0$, l'équation de la droite qui passe par les points de contact sera $A(mn' - m'n)x + \dots = 0$. Donc: si ces deux droites touchent la conique $U + (\alpha x + \beta y + \gamma z)^2 = 0$, l'équation de la droite qui passe par les deux points de contact sera

$$A(mn' - m'n)x + \dots + (\alpha x + \beta y + \gamma z)[\alpha(mn' - m'n) + \beta(nl' - n'l) + \gamma(lm' - l'm)] = 0 \quad \dots\dots (13).$$

Les formules obtenues seront utiles pour la solution du problème du mémoire suivant, [91]. Je les ai rapprochées ici pour ne pas interrompre cette solution.

[91] Sur le problème des contacts

[Book p. 522]

[From the *Journal für die reine und angewandte Mathematik (Crelle)*, tom. XXXIX. (1850), pp. 4–13.]

Je me propose ici la solution analytique du problème suivant:

“Étant données trois coniques inscrites à une même conique: trouver une autre conique, aussi inscrite à cette conique, qui touche les trois coniques inscrites; et tirer de là les constructions géométriques ordinaires.”

Je commence par récapituler quelques-unes des propriétés d'un système de trois coniques inscrites à la même conique.

Un système de six droites qui passent trois à trois par quatre points, s'appelle *quadrangle*. Les points de rencontre des côtés opposés sont les *centres* du quadrangle; les côtés du triangle formé par ces trois centres sont les *axes* du quadrangle. De même, un système de six points situés trois à trois sur quatre

droites, s'appelle *quadrilatère*. Les droites qui passent par les angles opposés sont les *axes*; et les angles du triangle formé par les trois axes sont les *centres* du quadrilatère.

Deux coniques quelconques se coupent en quatre points qui forment un *quadrangle* inscrit aux deux coniques. Elles ont quatre tangentes communes qui forment un *quadrilatère* circonscrit aux deux coniques. Le quadrangle inscrit et le quadrilatère circonscrit ont les mêmes centres et les mêmes axes.

Si deux coniques sont circonscrites ou inscrites l'une à l'autre, la droite qui passe par les deux points de contact s'appelle *chorde de contact*, et le point de rencontre des deux tangentes communes s'appelle *centre de contact*.

Cela posé: les coniques circonscrites à deux coniques données, peuvent être divisées en trois classes: une conique circonscrite appartient à une quelconque de ces trois classes, selon que les points de rencontre des cordes de contact de la conique

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circonscrite et de chacune des deux coniques données coïncide avec un quelconque des trois centres du quadrangle inscrit, ou du quadrilatère circonscrit; ou, si l'on veut, selon que la droite qui passe par les centres de contact de la conique circonscrite et des deux coniques données, coïncide avec un quelconque des trois axes du quadrangle inscrit, ou du quadrilatère circonscrit.

En considérant les deux coniques données, et une conique circonscrite, nous dirons que les deux côtés du quadrangle inscrit qui se coupent dans le point d'intersection des deux cordes de contact, sont les *axes de symptose* des deux coniques données, et que les deux angles du quadrilatère circonscrit, situés sur la droite qui passe par les deux centres de contact, sont les *centres d'homologie* des deux coniques données.

Soient maintenant inscrites trois coniques à la même conique. En combinant deux à deux ces trois coniques, les six axes de symptose se couperont trois à trois en quatre points que nous nommerons *centres de symptose*, et les six centres d'homologie seront situés trois à trois sur quatre droites que nous appellerons *axes d'homologie*.

Soient

$$U + V_1^2 = 0, \quad U + V_2^2 = 0, \quad U + V_3^2 = 0$$

les équations des trois coniques inscrites, ou

$$\begin{aligned} U &= Ax^2 + By^2 + Cz^2 + 2Fyz + 2Gxz + 2Hxy, \\ V_1 &= \alpha_1 x + \beta_1 y + \gamma_1 z, \\ V_2 &= \alpha_2 x + \beta_2 y + \gamma_2 z, \\ V_3 &= \alpha_3 x + \beta_3 y + \gamma_3 z. \end{aligned}$$

Si

$$lx + my + nz = 0$$

est l'équation d'une tangente commune aux coniques $U + V_1^2 = 0$, $U + V_2^2 = 0$, les formules de la "Note sur quelques formules &c." [90], en adoptant la notation de cette note, donneront les équations

$$\begin{aligned} (K + \mathfrak{A}\alpha_1^2 + \dots)(\mathfrak{A}l^2 + \dots) - (\mathfrak{A}\alpha_1 l + \dots)^2 &= 0, \\ (K + \mathfrak{A}\alpha_2^2 + \dots)(\mathfrak{A}l^2 + \dots) - (\mathfrak{A}\alpha_2 l + \dots)^2 &= 0, \end{aligned}$$

qui serviront à déterminer les valeurs de l , m , n ; on obtient par là l'expression

$$\sqrt{(K + \mathfrak{A}\alpha_1^2 + \dots)}(\mathfrak{A}\alpha_2 l + \dots) - \sqrt{(K + \mathfrak{A}\alpha_2^2 + \dots)}(\mathfrak{A}\alpha_1 l + \dots),$$

qui fait voir que l'équation $lx + my + nz = 0$ est satisfaite en écrivant

$$\sqrt{(K + \mathfrak{A}\alpha_1^2 + \dots)}(\mathfrak{E}\alpha_2 + \mathfrak{F}\beta_2 + \mathfrak{C}\gamma_2) - \sqrt{(K + \mathfrak{A}\alpha_2^2 + \dots)}(\mathfrak{E}\alpha_1 + \mathfrak{F}\beta_1 + \mathfrak{C}\gamma_1).$$